POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Artificial Intelligence

Semester: Spring

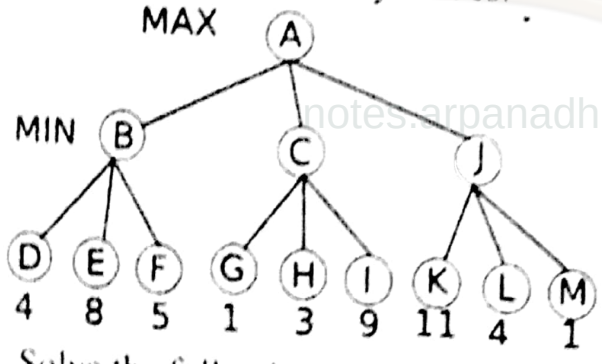
Year : 2012
Full Marks : 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) What is AI? Explain various application areas of AI. 7
   b) What is state space search? Give any example of a game which happens to be a problem of state space search and justify your answer properly. 8
2. a) Define Problem formulization. What are the components of problem formulization? Formulize 8-Puzzle problem. 8
   b) Compare Breadth First Search and Depth First Search in terms of time and space complexity, completeness and optimality. 7
3. a) In the figure below, use minimax search with alpha-beta pruning to decide the next move (node) by the MAX player from node A. The numbers indicate utility values. 8



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   b) Solve the following crypto arithmetic problem. 7

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4. a) Assume the following facts:
   i) Bhaskar is a physician
   ii) All physician knows surgery.
   iii) All MBBS are physician.
   Prove that "Bhaskar knows surgery" using resolution.

   b) Covert the following into WFfx:
   i) Everyone is liked by someone.
   ii) It is 2012.
   iii) Ravi's father is Rani's father.
   iv) All cats have tail and whiskers.

5. a) Define WFF. State and explain inference rules used in FOPL.
   b) Contrast and compare Semantic Nets and Frames for knowledge representation.

6. a) What is learning? Explain rote learning and induction learning. Why the process of knowledge acquisition so tedious.
   b) Define supervised & unsupervised learning. How perception is used to training the neural network.

7. Write short notes on **any two:**
   a) Turing Test of Intelligence
   b) Tautology, contradiction and contingency
   c) Expert System

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